

Jiayi Chen

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Shantou University, Shantou, China

EDUCATION

Shantou University

Sep. 2024 – Jun. 2028

Bachelor of Engineering

- **Major:** Data Science and Big Data Technology
- **GPA:** 3.97/5.0 (top 10%) **Average Score:** 89.7/100
- **Relevant Coursework:** Data Structures, Advanced Mathematics, Computer Organization, Probability & Mathematical Statistics, Database Systems, Object-Oriented Programming, LLM Agent Theory & Practice, Multi-Source Information Fusion

MANUSCRIPTS

- [1] **Jiayi Chen**, Zichun Lin, Dan Yang, et al. “Structural-Evidence Perturbation Learning for Hyperedge Prediction” *The 41st Annual AAAI Conference on Artificial Intelligence (AAAI’27, CCF-A)*. (Under Review)
- [2] **Jiayi Chen**, Jiahui Zhong, Zhifeng Hao, et al. “SPD: Side-Path Decoupled Adaptation for Few-Shot Motor Imagery EEG Decoding” *The IEEE International Conference on Bioinformatics and Biomedicine (BIBM’26, CCF-B)*. (Under Review)
- [3] Zichun Lin, **Jiayi Chen**, Dan Yang, et al. “Dual-View Fuzzy Membership and Candidate-Adaptive Multi-Source Evidence Fusion for Hyperedge Prediction” *The Fifteenth International Conference on Learning Representations (ICLR’27, CCF-A)*. (Under Review)

PATENTS

- [1] Jiajun Xian, Li Qi, Can Zhou, **Jiayi Chen**, et al. A Multi-Class Classification Method for Medical Data Based on Intuition-Driven Fuzzy Least Squares, Invention Patent, 202610640838.X, Preliminary Review Passed.

RESEARCH EXPERIENCE

Structural-Evidence Perturbation for Hyperedge Prediction | Project Lead

Mar. 2026 – Jul. 2026

- Developed **SEP-HP**, a structural-evidence perturbation framework that addresses hard negatives indistinguishable by static overlap and co-occurrence.
- Designed **SEDR** with member- and supporting-hyperedge deletion responses to model candidate-specific structural dependencies and learn residual corrections beyond static scores.
- Integrated **risk-aware gated refinement** and validated the framework across six hypergraph benchmarks and four negative-sampling protocols, achieving the best average AUROC/AP on all six datasets.

Side-Path Decoupled Adaptation for MI-EEG Decoding | Project Lead

May. 2026 – Jul. 2026

- Improved few-shot MI EEG decoding by designing SPD, a side-path adapter that preserves main route’s pretrained decision route during 30% adaptation
- Controlled feature correction by using a shallow convolutional side encoder, zero-initialized residual injection, stop-gradient alignment, and task auxiliary loss
- Delivered a lightweight adaptation pipeline that improved mean decoding accuracy across three public MI datasets with only 2.13% additional parameters

COVID-19 Excess Disease Burden Forecasting Based on GBD Data | Research Group Member

Sep. 2025 – Mar. 2026

- Constructed GBD-based longitudinal datasets covering incidence, mortality, and DALY indicators, with standardized preprocessing for cross-year disease burden analysis
- Developed a hybrid counterfactual forecasting model combining trend extrapolation and nonlinear residual modeling to estimate expected burden without the COVID-19 shock
- Compared predicted counterfactual trajectories with observed 2020–2023 data to quantify excess disease burden associated with the pandemic

HONOR & AWARDS

National Encouragement Scholarship

Nov. 2025

Special Talent Scholarship, Shantou University

Dec. 2024

Third Prize Comprehensive Scholarship, Shantou University

Nov. 2025

Third Prize, 2025 CUMCM (Guangdong Province)

Oct. 2025

SKILLS

- **Languages:** English (fluent); Cantonese & Mandarin Chinese (native)
- **Professional Skills:** Python, C/C++, MS Office, MATLAB, data processing, etc.
- **Design & Editing:** Proficient in Xiumi Editor for graphic design, layout, and visual content production
- **Interests:** Travel and photography; strong aesthetic sensibility and visual composition skills